

ARIN 5203 Assignment3: Generative Modeling (100 points)

NOTE: You only need to complete **Section 1** (Generative Adversarial Networks) plus **ONE** of the following three sections: either Section 2 (Variational Autoencoders), Section 3 (Diffusion Models), OR Section 4 (Flow Matching Models). You do **not** need any additional packages and using them is **strongly discouraged**.

1 Generative Adversarial Networks (60 points)

We will be training Generative Adversarial Networks (GAN) on the CUB 2011 Dataset.

- **Setup:** Run the following command to set up everything you need for the assignment:
`./setup.sh /path/to/python_env/lib/python3.8/site-packages`. Please use the PyTorch installation from the previous homework.
- **Question:** Follow the instructions in the README.md file in the `gan/` folder to complete the implementation of GANs
- **Debugging Tips:**
 - GAN losses are pretty much meaningless! If you want to understand if your network is learning, visualize the samples. The FID score should generally be going down as well
 - Do NOT change the hyperparameters at all, they have been carefully tuned to ensure the networks will train stably. If things aren't working, it's a bug in your code.
 - For debugging, disable JIT using `export PYTORCH_JIT=0 python ...` and disable AMP by using the flag `- -disable_amp`. However, do note that disabling JIT will cause the FID calculation to fail. So only disable JIT to make sure that your network code runs correctly, then re-enable when training. If you observe any errors involving type mismatches and tensors that have half types, it is due to AMP. You may need to explicitly cast the tensor using `.half()`.
 - Here is a sample image from WGAN-GP at the end of training. The other networks may have variations, but they should look similar.



• **Expected Results:**

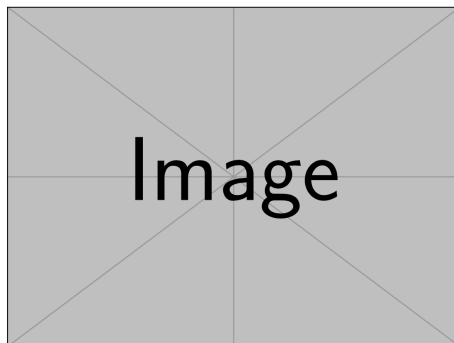
- Vanilla GAN: Final FID should be less than 110.
- LS-GAN: Final FID should be less than 90.
- WGAN-GP: Final FID should be less than 70.

- **Deliverables:** The code will log plots to `gan/data_gan`, `gan/data_ls_gan`, and `gan/data_wgan_gp`. Extract plots and paste them into the appropriate section below. Note for all questions, we ask for the final FID. Final FID is computed using 50K samples, at the very end of training. See the final printout for “Final FID (Full 50K).”

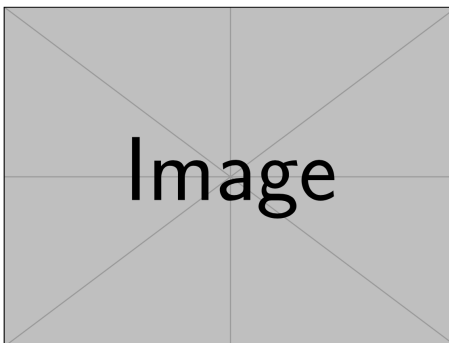
1. Paste your plot of the samples and latent space interpolations from Vanilla GAN, as well as the final FID score you obtained.

Solution:

Vanilla GAN FID:

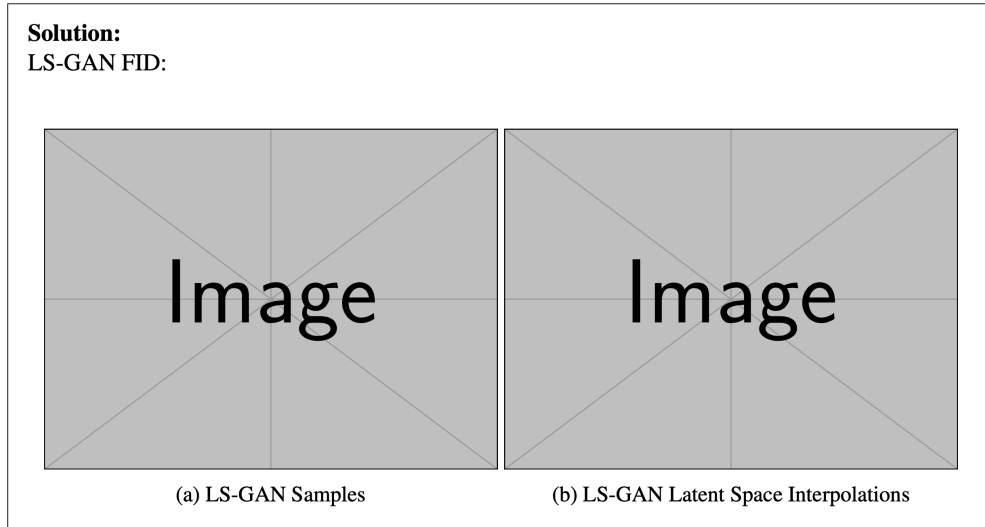


(a) Vanilla GAN Samples

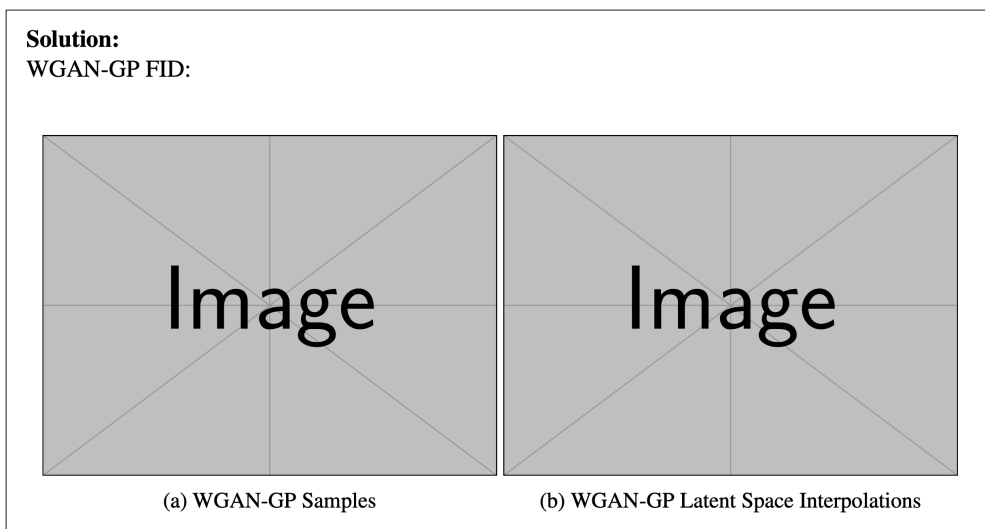


(b) Vanilla GAN Latent Space Interpolations

2. Paste your plot of the samples and latent space interpolations from LS-GAN, as well as the final FID score you obtained.



3. Paste your plot of the samples and latent space interpolations from WGAN-GP, as well as the final FID score you obtained.



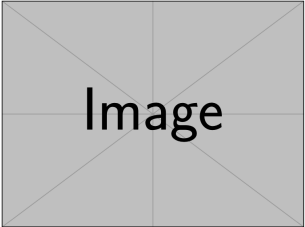
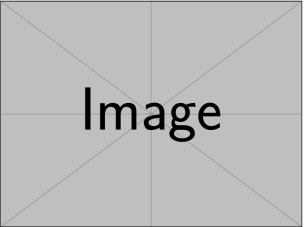
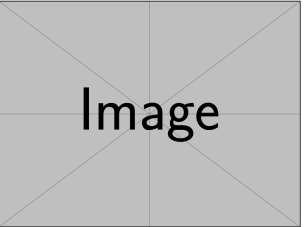
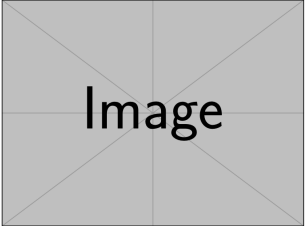
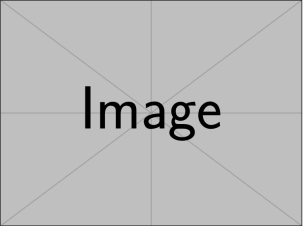
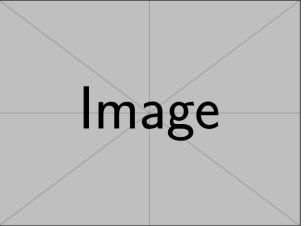
2 Variational Autoencoders (40 pts)

We will be training Autoencoders and Variational Auto-Encoders (VAE) on the CIFAR-10 dataset.

- **Question:** Follow the instructions in the README.md file in the vae/ folder to complete the implementation of VAEs.
- **Debugging Tips:**
 - Make sure the auto-encoder can produce good-quality reconstructions before moving on to the VAE. While the VAE reconstructions might not be clear and the VAE samples even less so, the auto-encoder reconstructions should be very clear.
 - If you are struggling to get the VAE portion working: debug the KL loss independently of the reconstruction loss to ensure the learned distribution matches the standard normal.
- **Expected Results:**
 - AE: reconstruction loss should be < 40 , reconstructions should look similar to the original image.
 - VAE: reconstruction loss should be < 145 ($\beta = 1$ case).
 - VAE: reconstruction loss should be < 125 when annealing β .
- **Deliverables:** The code will log plots to different folders in vae. Please paste the plots into the appropriate place for the questions below. Note for ALL questions, use the reconstructions and samples from the final epoch (epoch 19).

1. Autoencoder: For each latent size, paste your plot of the reconstruction loss curve and reconstructions.

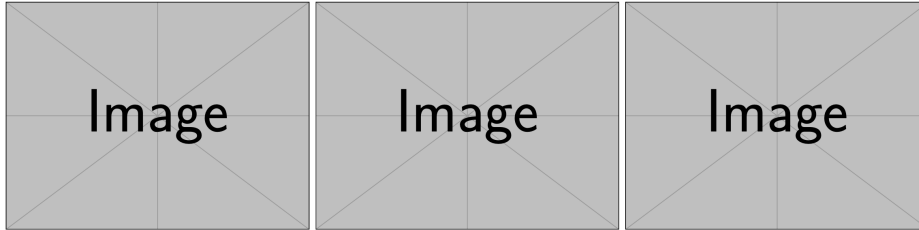
Solution:

		
(a) Loss (size 16)	(b) Loss: (size 128)	(c) Loss: (size 1024)
		
(d) Reconstructions: (size 16)	(e) Reconstructions: (size 128)	(f) Reconstructions: (size 1024)

2. VAE: Choose the β that results in the best sample quality, β^* . Report the β^* you used in your experiments. In addition, paste the reconstruction, KL loss curve plots, and the sample images corresponding to the VAE trained using constant β^* and the VAE trained using β annealing scheme with β^* .

Solution:

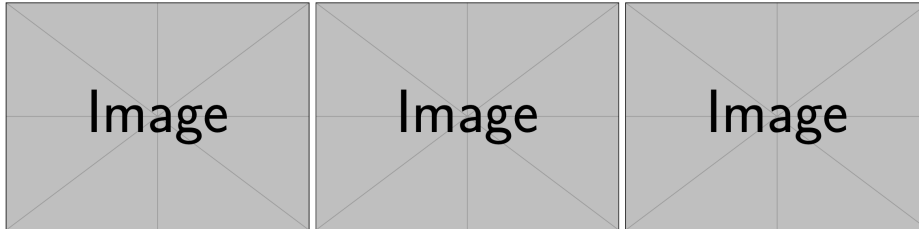
β^* value:



(a) Recon. Loss: constant β^*

(b) KL Loss: constant β^*

(c) Samples: constant β^*



(d) Recon. Loss: β annealed

(e) KL Loss: β annealed

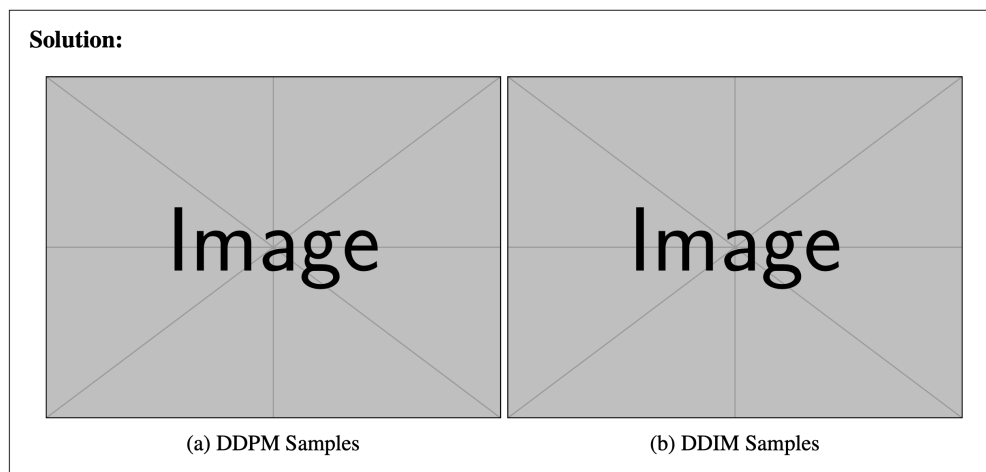
(f) Samples: β annealed

3 Diffusion Models (40 points)

We will be running inference using a pre-trained diffusion model (DDPM) on CIFAR-10.

- **Setup:** Download our pre-trained checkpoint for DDPM from [here](#).
- **Question:** Follow the instructions in the README.md file in the diffusion/ folder to complete the implementation of the sampling procedures for Diffusion Models.
- **Expected Results:**
 - FID of less than 60 for DDPM and DDIM
- **Deliverables:** The code will log plots to diffusion/data_ddpm and diffusion/data_ddim. Extract plots and paste them into the appropriate section below.

1. Paste your plots of the DDPM and DDIM samples..



2. Paste in the FID score you obtained from running inference using DDPM and DDIM.

Solution:

- DDPM FID:
- DDIM FID:

3. Compare the generated samples from your VAE and diffusion models. What do you observe? Does one generate better pictures than the other? Does the quality of the generated images match the FID differences?.

Solution:

4 Flow Matching Models (40 points)

We will be running inference using a pre-trained flow matching model (flow matching) on CIFAR-10.

- **Setup:** Download our pre-trained checkpoint for the flow matching model from [here](#).
- **Question:** Follow the instructions in the README.md file in the flow matching/ folder to complete the implementation of the sampling procedures for Flow Matching Models.
- **Expected Results:**
 - FID of less than 60 for Euler ODE sampling.
- **Deliverables:** The code will log plots to flow matching/flow_euler. Extract the plot and paste it into the section below..

1. Paste your plots of the Euler ODE samples.

Solution:

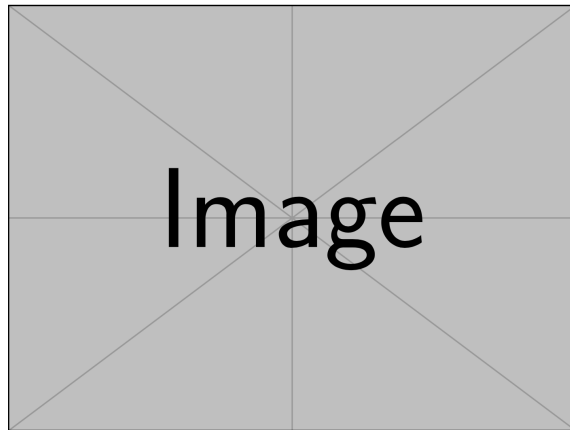


Figure 4.1: Flow Matching Euler ODE Samples

2. Paste in the FID score you obtained from running inference using Euler ODE sampling.

Solution:

- Flow Matching FID:

3. Compare the generated samples from your VAE and flow matching models. What do you observe? Does one generate better pictures than the other?

Solution:

5 Submission Instructions

You will submit this assignment on Canvas as two submissions - one for **Assignment3 [coding]** and another for **Assignment 3 [written]**:

- (a) Run the `collect_submission.sh` script to produce your `assignment3.zip` file.
- (b) Add all relevant plots and text answers in the written solutions.
- (c) Upload your `assignment3.zip` and your written solutions to Canvas.